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Studierichting: Informatica
Oefeningenreeks 4A nr. 31.12

$$\int \frac{3xdx}{1+\sqrt{1+2x}}$$
$$= 3\int \frac{xdx}{1+\sqrt{1+2x}}$$
Stel $t = 2x+1 \Rightarrow dt = 2dx$
$$= \frac{1}{4}\int \frac{t-1}{\sqrt{t}+1}dt$$
Stel $u = \sqrt{t}+1 \Rightarrow du = \frac{1}{2\sqrt{t}}dt$
$$= \frac{1}{4}\left(2\int (u-2)(u-1)du\right)$$
$$= \frac{2}{4}\left(\int u^2du - 3\int udu + 2\int du\right)$$
$$= \frac{2}{4}\left(\frac{u^3}{3} - \frac{3u^2}{2} + 2u\right) + \text{C}$$
$$= \frac{1}{4}\left(\frac{2u^3}{3} - 3u^2 + 4u\right) + \text{C}$$
$$= \frac{1}{4}\left(\frac{2(\sqrt{t}+1)^3}{3} - 3(\sqrt{t}+1)^2 + 4(\sqrt{t}+1)\right) + \text{C}$$
$$= \frac{1}{4}\left(\frac{2(\sqrt{2x+1}+1)^3}{3} - 3(\sqrt{2x+1}+1)^2 + 4(\sqrt{2x+1}+1)\right) + \text{C}$$
$$= \frac{\sqrt{2x+1}+1}{6} - \frac{3(\sqrt{2x+1}+1)^2}{4} + \sqrt{2x+1}+1 + \text{C}$$
$$= \frac{(2x+1)^{\frac{3}{2}} - 3x}{2} + \text{C}$$

References